

All-but-one MMS Allocation for Chores*

Jiawei Qiu
IOTSC,
University of Macau
Macau, China
mc25118@um.edu.mo

Xiaowei Wu
IOTSC,
University of Macau
Macau, China
xiaoweiwu@um.edu.mo

Cong Zhang[†]
IOTSC,
University of Macau
Macau, China
yc27429@um.edu.mo

Shengwei Zhou
SPMS, Nanyang
Technological University
Singapore
shengwei.zhou@ntu.edu.sg

Abstract

We study the problem of fairly allocating m indivisible chores among n agents with additive cost functions. While the maximin share (MMS) is a prominent fairness criterion in theory, exact MMS allocations do not always exist. This has motivated relaxation that guarantees MMS fairness for only a subset of agents, aiming to maximize the number of satisfied agents. However, for chore allocation, guaranteeing most agents their full MMS is trivial but highly unsatisfactory, e.g., overburdening a single agent, which is undesirable in real-world platforms aiming for user retention and satisfaction. To address this, we propose a stronger and more practical notion called α -approximate all-but-one MMS (α -AMMS), which guarantees that $n - 1$ agents receive their full MMS value, while the remaining agent receives an α -approximation. This model reflects common platform design goals, where satisfying the vast majority of users is critical, and near-fairness for the rest is acceptable. We show that there exist α -AMMS allocations, with $\alpha = 9/8$ for three agents; $\alpha = 4/3$ for four agents; and $\alpha = (n + 1)^2/4n$ for $n \geq 5$ agents.

CCS Concepts

• **Theory of computation** → **Algorithmic game theory**; *Approximation algorithms analysis*; Scheduling algorithms.

Keywords

fair allocation, maximin share, multi-agent

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1 Introduction

The fair allocation of resources or tasks among multiple agents is a fundamental problem in multi-agent systems, with profound implications for online platforms, digital marketplaces, and human

computation systems. From allocating tasks in crowd-sourcing platforms like Amazon Mechanical Turk to distributing computational jobs in distributed systems, the challenge of achieving fairness while maintaining efficiency is ubiquitous. Since its inception by Steinhaus [45] in 1948, the fair division problem has been studied extensively in computer science, economics, and social choice theory. This line of research can be categorized into two settings: the allocation of goods (e.g., rewards, resources, or items of value) and the allocation of chores (e.g., tasks, jobs, or undesirable items). In this work, we focus on the latter, which naturally arises in online task assignment, load balancing, and human-in-the-loop systems where tasks must be distributed fairly among participants (while some of our results can be applied to the case of goods). For each agent $i \in N$, we use $c_i : 2^M \rightarrow \mathbb{R}^+ \cup \{0\}$ to denote her cost function, which is additive, i.e., for any bundle $S \subseteq M$, $c_i(S) = \sum_{e \in S} c_i(e)$. We consider indivisible items, where each chore needs to be allocated to exactly one agent. When items are divisible, a natural criterion is *proportionality* [45] that requires each agent to receive at most her proportional share of all chores. Since proportionality is not guaranteed to exist for indivisible items (consider allocating one item to two agents), some refined criteria have received a lot of attention. Among all the notions, the maximin share (MMS) is one of the most popular criteria, since introduced by Budish [19]. For each agent i , her maximin share MMS_i is defined as the minimum cost she can guarantee if she divides all the items into n bundles and picks the worst bundle. When there are only two agents, the definition of MMS coincides with the well-studied mechanism *divide-and-choose*. Unfortunately, when there are more than two agents, MMS allocations are not guaranteed to exist for both goods and chores [11, 24, 41]. Much research tends to consider relaxations, such as the approximation of MMS, where an allocation is α -MMS if every agent receives a bundle with cost at most α times her MMS.

Approximate MMS. The very first upper bound of $2 - 1/n$ for the approximation ratio is shown by Aziz *et al.* [11], where the algorithm only requires the ordinal preferences of the agents on the items. The result is further improved to $5/3$ by Aziz *et al.* [10] and then to $3/2$ by Feige and Huang [22]. With cardinal information, Barman and Krishnamurthy [16] showed the existence of $4/3$ approximations of MMS, which was further improved to $11/9$ by Huang and Lu [36], and then to $13/11$ by Huang and Segal-Halevi [37]. Regarding the lower bounds, Feige *et al.* [24] showed that no algorithm can achieve approximation ratios smaller than $44/43$ for the allocation of chores, even for three agents with cardinal information. For the analogous problem of goods, there is a sequence of works on the approximation of MMS [3, 4, 16, 28–30, 33, 41], which leads to the state-of-the-art ratio of $7/9$ [38].

The above works focused on guaranteeing all agents a relaxed threshold (rather than MMS). For the allocation of goods, there is a

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[†]Corresponding author



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line of research focusing on guaranteeing a fraction of agents their MMS. The goal is to investigate the maximum fraction of agents that can be guaranteed. To evaluate the allocation, Hosseini and Searns [34] proposed the framework of (α, β) -MMS that guarantees α -fraction of the agents a β -approximation of their MMS. For the remaining $(1 - \alpha)$ -fraction of agents, there is no guarantee. They showed the existence of $(\frac{n-1}{n}, 1)$ -MMS allocations for $n \leq 4$ agents, and the existence of $(\frac{2}{3}, 1)$ -MMS allocations for $n \geq 5$. When applying the (α, β) -MMS framework to the chores setting, guaranteeing most agents their MMS value is trivial, e.g., by allocating all chores to one single agent. While such an allocation is $(\frac{n-1}{n}, 1)$ -MMS, the allocation outcome is highly undesired. Therefore, we propose to study a stronger version of $(\frac{n-1}{n}, 1)$ -MMS, and introduce a framework called *all-but-one maximin share* (AMMS). We call an allocation α -AMMS if there are $n - 1$ agents guaranteed their MMS and the remaining agent is guaranteed an α -approximation of her MMS, where $\alpha \geq 1$. A natural question to ask is

When ensuring MMS for $n - 1$ agents, what is the smallest approximation of MMS we can guarantee for the remaining agent?

By definition, every α -AMMS allocation is also α -MMS. Consequently, the established lower bound of $44/43$ [24] for α -MMS allocations directly applies to α -AMMS, even in the case of three agents. Prior to our work, however, no non-trivial upper bounds on the achievable α had been established for α -AMMS.

1.1 Our Results

In this paper, we consider the existence of α -AMMS allocations for indivisible chores. We first consider the case of three agents, and show the existence of $9/8$ -AMMS allocations. Then we extend our result to instances with four agents and show the existence of $\frac{4}{3}$ -AMMS allocations.

Result 1 (Theorem 3.1). *For any instance with three agents, there exist $9/8$ -AMMS allocations.*

Result 2 (Theorem 4.1). *For any instance with four agents, there exist $4/3$ -AMMS allocations.*

Our analysis follows the popular matching-based framework for computing fair allocations [2, 18, 25, 27, 39]. For the case of three agents, we first partition the items into three bundles B_1, B_2, B_3 based on the MMS partition of agent 1. Then we construct a bipartite graph between the three agents and the three bundles, where we put an edge between agent i and bundle B if and only if $c_i(B) \leq \text{MMS}_i$, where MMS_i denotes the MMS of agent i . Note that agent 1 has an edge to every bundle. Clearly, if the graph admits a perfect matching, then a 1-AMMS allocation is found. Otherwise, we check whether there exists a bundle B with cost at most $9/8 \cdot \text{MMS}_i$, such that after assigning bundle B to agent i , the remaining graph admits a perfect matching. If neither case happens, then we assign the bundle B that agent 2 dislikes the most to agent 1, and run a *divide-and-choose* procedure between agents 2 and 3. The main challenge is that assigning bundle B to agent 1 may cause an increase in MMS of agent 2. Our main technical contribution is showing that such an assignment does not increase the MMS of agent 2 by a factor larger than $9/8$, utilizing the fact that B is the bundle agent 2 dislikes the most, and the structural properties of MMS. We show that our analysis for three agents is tight by providing an instance for which our algorithm returns a $9/8$ -AMMS allocation. We also show that

similar analysis can be extended to the allocation of goods to show the existence of $6/7$ -AMMS allocations (see Appendix B).

We further extend the analysis framework to the case of four agents for the allocation of chores, showing the existence of $4/3$ -AMMS allocations (Result 2). Part of our analysis is inspired by the *atomic bundle* framework [23, 24] which involves intersections of bundles from different agents' MMS partitions.

Finally, we consider instances with a general number of agents. We present an upper bound of $(n + 1)^2/4n$ for the approximation ratio of AMMS allocations (in Appendix A). Our analysis is based on a matching-based framework and involves a reduction from general instances to smaller instances that preserve some nice properties of the original instance.

Result 3 (Theorem A.1). *For any instance with $n \geq 5$ agents, there exist $\frac{(n+1)^2}{4n}$ -AMMS allocations.*

1.2 Related Works

Since the notion of MMS is very well-studied and there is a vast literature, here we only review the most closely related works. For the allocation of chores, Feige and Norkin [23] show that $19/18$ -MMS allocations always exist for three agents. Their algorithm guarantees that one agent achieves her proportional share, while the other two agents achieve their $19/18$ -MMS. Thus, this result can not imply the existence of $19/18$ -AMMS allocations for three agents. For four to seven agents, $20/17$ -MMS allocations are guaranteed to exist [37]. For the allocation of goods, Feige and Norkin [23] showed the existence of $11/12$ -MMS allocations for three agents, and established an upper bound of $39/40$. For four agents, $4/5$ -MMS allocations are guaranteed to exist [13, 30]. Apart from the cardinal approximation, several works focused on the ordinal approximation of MMS, called 1-out-of- d MMS [1, 34, 35], where when we define the maximin share, we ask the agent to partition the items into d (instead of n) bundles. The current state-of-the-art results are 1-out-of- $\lfloor \frac{9n}{11} \rfloor$ MMS for chores [26] and 1-out-of- $\lfloor \frac{4n}{3} \rfloor$ for goods [5].

MMS allocations have also been studied in some generalized settings, e.g., beyond additive functions [6, 31, 43, 44, 46]. Zhou *et al.* [48] and Kulkarni *et al.* [40] studied (approximate) MMS allocations in the online setting with a normalization assumption. Additionally, there is a line of literature focusing on allocating items to strategic agents while guaranteeing (approximate) MMS allocations [7, 8, 10, 15]. For asymmetric agents, MMS inspired several fairness notions such as weighted MMS (WMMS) [9, 21, 42, 47] and AnyPrice Share (APS) [12, 22]. Furthermore, Pairwise MMS [20] and Groupwise MMS [14] are defined to consider the fairness between pairwise or groupwise agents.

Recently, Akrami *et al.* [5] proposed a framework called T -MMS that guarantees different approximation ratios for different agents. For a non-increasing sequence $T = (\tau_1, \dots, \tau_n)$ of numbers, if the agent at rank i in the order gets a bundle of value at least τ_i times her MMS value, the allocation is said to be a T -MMS allocation. For the allocation of goods, Akrami *et al.* showed that there exist T -MMS allocations where $\tau_i \geq \max\{\frac{3}{4} + \frac{1}{12n}, \frac{2n}{2n+i-1}\}$ for all $i \in N$.

2 Preliminary

We consider allocating a set of m indivisible chores to n agents. We use $M = \{e_1, \dots, e_m\}$ to denote the set of items and $N = \{1, \dots, n\}$

to denote the set of agents, respectively. Each agent $i \in N$ has an additive cost function $c_i : 2^M \rightarrow \mathbb{R}^+ \cup \{0\}$. An instance is denoted by $\mathcal{I} = (M, N, \mathbf{c})$, where $\mathbf{c} = (c_1, \dots, c_n)$ is the set of cost functions. A k -partition of items M is denoted by $\mathbf{P} = \{P_1, \dots, P_k\}$ where $P_i \cap P_j = \emptyset$ for all $i \neq j$ and $\cup_{1 \leq i \leq k} P_i = M$. Specifically, we call an n -partition an allocation $\mathbf{X} = (X_1, \dots, X_n)$ if X_i is the bundle allocated to agent i . For convenience, we use $X_i + e$ to denote $X_i \cup \{e\}$ and $X_i - e$ to denote $X_i \setminus \{e\}$. We use $[k]$ to denote $\{1, \dots, k\}$. Given any integer k , let $\Pi_k(M)$ be the set of all k -partitions of M .

DEFINITION 2.1 (PROPORTIONALITY). Given an instance $\mathcal{I} = (M, N, \mathbf{c})$, for any $i \in N$ the proportional share of agent i is $\frac{1}{n} \cdot c_i(M)$. An allocation is said to satisfy proportionality (PROP) if for every agent $i \in N$, $c_i(X_i) \leq \frac{1}{n} \cdot c_i(M)$.

DEFINITION 2.2. Given a set of items $M' \subseteq M$, an integer $k \leq n$, the (M', k) -maximin share¹ of agent i is defined as follows:

$$\text{MMS}_i(M', k) = \min_{\mathbf{P} \in \Pi_k(M')} \max_{j \in [k]} \{c_i(P_j)\}.$$

An n -partition of M is called an MMS partition for agent i if all the bundles in this partition have a cost at most $\text{MMS}_i(M, n)$.

DEFINITION 2.3 (α -MMS). An allocation $\mathbf{X}$ is said to satisfy α -approximate maximin share for some $\alpha \geq 1$, if for every agent $i \in N$,

$$c_i(X_i) \leq \alpha \cdot \text{MMS}_i(M, n).$$

When $\alpha = 1$, we call the allocation an MMS allocation. For convenience, we simply write MMS_i when M and n are clear from the context. Without loss of generality, for any agent $i \in N$ we normalize c_i such that $\text{MMS}_i = 1$. A bundle X is called MMS-feasible for agent i if $c_i(X) \leq 1$. We may also simply say that agent i likes bundle X if it is MMS-feasible for agent i . Given any partition $\mathbf{P} = \{P_1, \dots, P_k\}$ of items $M' \subseteq M$, we say $\mathbf{P}$ is a $(\alpha, 1, \dots, 1)$ -partition for agent i if there exists a bundle P_j such that $c_i(P_j) \leq \alpha$ and $c_i(P_l) \leq 1$ for any $l \neq j$. As a common observation of MMS, we have $c_i(M) \leq n$.² Furthermore, for any instance, we have $c_i(e) \leq 1$ for all $e \in M$ according to the definition of MMS. Next, we introduce the framework of all-but-one maximin share (AMMS). We guarantee almost all agents ($n-1$ agents) an MMS allocation and the remaining one agent α times her MMS.

DEFINITION 2.4 (α -AMMS). An allocation $\mathbf{X}$ is α -approximate all-but-one maximin share (α -AMMS), if there exists an agent $i \in N$ with $c_i(X_i) \leq \alpha$ and $c_j(X_j) \leq 1$ holds for every agent $j \in N \setminus \{i\}$.

When $\alpha = 1$, the AMMS allocation is equivalent to the MMS allocation. Bouveret and Lemaître [17] showed that the *divide-and-choose* procedure returns an MMS allocation for two agents. Hence, in the following, we focus on the existence of α -AMMS for $n \geq 3$.

DEFINITION 2.5 (MMS-FEASIBILITY GRAPH). Given a partition $\mathbf{P} = \{P_1, \dots, P_n\}$, we construct a bipartite graph $G = (N \cup \mathbf{P}, E)$ in which we add an edge between agent $i \in N$ and bundle $P_j \in \mathbf{P}$ if bundle P_j is MMS-feasible for agent i , i.e., $E = \{(i, P_j) : i, j \in N, c_i(P_j) \leq 1\}$. For any set of agents $S \subseteq N$, we use $L(S)$ to denote the set of neighbors of S in the graph G .

¹Although the maximin share is defined through a min-max type formulation, if one considers chores as items with negative value, then the formulation is max-min.

²In any MMS partition $\mathbf{P}$ of agent i , we have $c_i(P_j) \leq 1$ which leads to $c_i(M) = \sum_{j \in N} c_i(P_j) \leq n$.

We remark that if $\mathbf{P}$ is a partition of M , we have $|L(i)| \geq 1$ for all $i \in N$. Otherwise, we have $\sum_{j=1}^n c_i(P_j) = c_i(M) > n$ which is a contradiction. Furthermore, given an MMS-feasibility graph G , if there exists a perfect matching in G , we can find an MMS allocation. By Hall's Theorem [32], we have the following observation.

OBSERVATION 2.6. For any MMS-feasibility graph G , there is no perfect matching for G if and only if there exists a subset $S \subseteq N$, such that $|S| > |L(S)|$.

Since we assumed that G does not admit a perfect matching, Observation 2.6 implies that there exist subsets $S \subseteq N$ with $|S| > |L(S)|$. We consider the largest subset S^* among them:

$$S^* = \operatorname{argmax}_{S \subseteq N: |S| > |L(S)|} \{|S|\}.$$

OBSERVATION 2.7. There exists a perfect matching between agents $N \setminus S^*$ and bundles $\mathbf{P} \setminus L(S^*)$.

PROOF. We show that for any subset $S' \subseteq N \setminus S^*$ we have $|S'| \leq |L(S')|$, which directly implies there exists a perfect matching between $N \setminus S^*$ and $\mathbf{P} \setminus L(S^*)$, by Hall's Theorem. Assume otherwise that there exists a subset $S' \subseteq N \setminus S^*$ such that $|S'| > |L(S')|$. Then we have $|L(S^* \cup S')| \leq |L(S^*)| + |L(S')| < |S^*| + |S'| = |S^* \cup S'|$, which is a contradiction towards that S^* is the maximum subset with $|S^*| > |L(S^*)|$. $\square$

See Figure 1 as an example: the graph (with four agents and four bundles) does not admit a perfect matching since $L(\{2, 3\}) = \{P_2\}$. Then we identify $S^* = \{1, 2, 3\}$. Clearly, there exists a perfect matching between agent $\{4\}$ and bundles $\{P_3, P_4\}$.

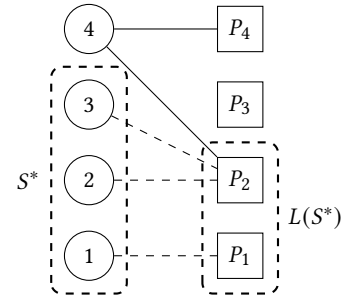


Figure 1: An example of an MMS-feasibility graph with $n = 4$.

Based on the MMS-feasibility graph, we introduce *valid* reductions and reduced instances. A reduction is valid if we can allocate some bundles to some agents, such that the proportionality of the remaining agents (on the remaining items) does not increase.

DEFINITION 2.8 (VALID REDUCTION). A reduction that allocates bundles $X_1, \dots, X_k$ to agents $1, \dots, k$ respectively is valid if and only if the following holds:

- $c_i(X_i) \leq 1$ for all $i \in [k]$, and
- $c_i(M \setminus (X_1 \cup \dots \cup X_k)) \leq n - k$ for all $i \in N \setminus [k]$.

DEFINITION 2.9 (REDUCED INSTANCE). Given an instance $\mathcal{I} = (M, N, \mathbf{c})$, a valid reduction outputs a new instance $\mathcal{I}' = (M', N', \mathbf{c}')$ such that $M' \subseteq M, N' \subseteq N$ and $c_i(M') \leq |N'|$ holds for all $i \in N'$. We call such an instance $\mathcal{I}'$ a reduced instance.

Let $I' = (M', N', c')$ be the reduced instance. Throughout this paper, each time we apply a reduction on an instance, we maintain that the reduction is valid: for any agent $i \in N'$, we guarantee his/her proportional share does not increase, i.e., $c_i(M') \leq |N'|$ for all $i \in N'$. However, it is possible that the MMS of an agents in the reduced instance is increased, i.e., $\text{MMS}_i(M', |N'|) > \text{MMS}_i$. Our main goal is to show that the increase is insignificant.

3 $\frac{9}{8}$ -AMMS for Three Agents

Given an instance $I = (M, N, c)$ with $n = 3$, we let $\mathbf{P} = \{P_1, P_2, P_3\}$ be an MMS partition for agent 3. Then we construct the MMS-feasibility graph $G = (N \cup \mathbf{P}, E)$. Note that we only have to consider the case when G does not admit a perfect matching. Otherwise, there exists an MMS allocation. Since $\mathbf{P}$ is an MMS partition for agent 3, all bundles are MMS-feasible for agent 3, i.e., $L(\{3\}) = \{P_1, P_2, P_3\}$. By Observation 2.6, for agents 1, 2 we have $|L(\{1, 2\})| = 1$, say P_1 . Next, we give an overview of the algorithm for computing $9/8$ -AMMS allocations.

The Algorithm. We examine each bundle in $\mathbf{P}$ to determine whether there exist two items e_1, e_2 in the same bundle with total cost larger than 1 for agent 1 or 2. If such items e_1, e_2 exist in any bundle, we assign the bundle containing e_1, e_2 to agent 3 and call the *divide-and-choose* procedure for the remaining items and agents. Otherwise (i.e., no such items exist), we assign the bundle with the highest cost with respect to (w.r.t.) c_1 to agent 3 and call the *load-balancing* procedure (based on c_1) for the remaining items and agents.

The main result of this section is stated as follows. The remainder of this section is dedicated to proving the theorem.

THEOREM 3.1. *For any instance $I = (M, N, c)$ with $n = 3$, there exist $\frac{9}{8}$ -AMMS allocations.*

Before we prove Theorem 3.1, we first show that assigning two items with a total cost larger than 1 (under c_1) to an agent does not increase the MMS of agent 1.

LEMMA 3.2. *Given an instance $I = (M, N, c)$, for any agent $i \in N$, if there exist two items e_1, e_2 with $c_i(e_1 + e_2) \geq 1$, then $\text{MMS}_i(M \setminus \{e_1, e_2\}, n - 1) \leq \text{MMS}_i(M, n)$.*

PROOF. Consider the MMS partition $\mathbf{P} = \{P_1, \dots, P_n\}$ for agent i , either there exists a bundle $P^* \in \mathbf{P}$ such that $\{e_1, e_2\} \subseteq P^*$ or there exist two bundles $P_1, P_2 \in \mathbf{P}$ such that $e_1 \in P_1, e_2 \in P_2$ and $P_1 \neq P_2$.

- For the case $\{e_1, e_2\} \subseteq P^*$, we must have $c_i(e_1 + e_2) = c_i(P^*) = 1$, since $c_i(e_1 + e_2) \geq 1 \geq \text{MMS}_i \geq c_i(P^*)$. Hence, we have $c_i(P^* \setminus \{e_1, e_2\}) = 0$ (which might be an empty set). Note that $\mathbf{P} \setminus \{P^*\}$ is a partition of items $M \setminus P^*$ with $c_i(P_j) \leq 1$ for all $P_j \in \mathbf{P} \setminus \{P^*\}$. Merging the set $P^* \setminus \{e_1, e_2\}$ with any bundle in $\mathbf{P} \setminus \{P^*\}$ will lead to a partition $\mathbf{P}'$ of $M \setminus \{e_1, e_2\}$ such that $c_i(P'_j) \leq 1$ for any $P'_j \in \mathbf{P}'$. Hence by the definition of MMS_i , we have $\text{MMS}_i(M \setminus \{e_1, e_2\}, n - 1) \leq \text{MMS}_i(M, n)$.
- For the case that $e_1 \in P_1, e_2 \in P_2$ such that $P_1 \neq P_2$, we have $c_i(P_1 \cup P_2) - c_i(e_1 + e_2) \leq 1$. Hence $\mathbf{P}' = (P_1 \cup P_2 \setminus \{e_1, e_2\}, P_3, \dots, P_n)$ is a partition of items $M \setminus \{e_1, e_2\}$ while $c_i(P_j) \leq 1$ for all $P_j \in \mathbf{P}'$. By the definition of MMS_i , we have $\text{MMS}_i(M \setminus \{e_1, e_2\}, n - 1) \leq 1$. $\square$

Next, we introduce the *divide-and-choose* and *load-balancing* procedures that we will use in the analysis. The *divide-and-choose*

procedure is described as follows. Given a set of items M' , a divider i , and a chooser j , we let agent i divide items M' into two bundles P_1, P_2 (with cost upper bounded by $\text{MMS}_i(M', 2)$) and let agent j choose a bundle in $\{P_1, P_2\}$ first.

OBSERVATION 3.3. *The divide-and-choose procedure guarantees that the divider receives a bundle with cost at most $\text{MMS}_i(M', 2)$ and the chooser receives a bundle with cost at most $\frac{1}{2} \cdot c_j(M')$.*

Algorithm 1: Divide-and-Choose(M', i, j)

Input: A set of items M' , a divider i and a chooser j .

1 Let agent i divide all items into two bundles P_1, P_2 ;

2 $X_j \leftarrow \text{argmin}_{B \in \{P_1, P_2\}} c_j(B)$; $X_i \leftarrow M' \setminus X_j$;

Output: An allocation $\mathbf{X} = (X_i, X_j)$.

Another well-studied procedure is called *load-balancing*. Given a set of items M' , an integer n' , and a cost function c_i , the algorithm allocates items in decreasing order of costs and assigns each item to the bundle with the minimum cost.

Algorithm 2: Load-Balancing(M', n', c_i)

Input: Items M' , partition size n' , and function c_i with $c_i(e_1) \geq c_i(e_2) \geq \dots \geq c_i(e_{|M'|})$.

1 For all $k \in [n']$, let $P_k \leftarrow \emptyset$;

2 **for** $j = 1, 2, \dots, |M'|$ **do**

3 Let $k^* \leftarrow \text{argmin}_{k \in [n']} c_i(P_k)$; // breaking ties arbitrarily

4 Update $P_{k^*} \leftarrow P_{k^*} + e_j$;

Output: A partition $\mathbf{P} = \{P_1, \dots, P_{n'}\}$.

Now we are ready to prove the existence of $9/8$ -AMMS allocations. Recall that we have an MMS-feasibility graph $G = (N \cup \mathbf{P}, E)$ where $\mathbf{P} = \{P_1, P_2, P_3\}$ is an MMS partition for agent 3. Moreover, it remains to consider the case that agents 1, 2 only like bundle P_1 . We consider whether there exist two items e_1, e_2 in one of the bundles of $\{P_2, P_3\}$ and an agent $i \in \{1, 2\}$ such that $c_i(e_1 + e_2) \geq 1$. If there exist such two items e_1, e_2 , then by Lemma 3.2 we have $\text{MMS}_i(M \setminus \{e_1, e_2\}, 2) \leq \text{MMS}_i(M, 3)$. We show that by using such a reduction, we find an MMS allocation by calling *divide-and-choose* procedure. If there do not exist such two items, we show that by allocating one of the bundles disliked by both agents 1 and 2 to agent 3 and repartitioning based on *load-balancing*, we can compute a $9/8$ -AMMS allocation.

LEMMA 3.4. *If there exist two items e_1, e_2 in some bundle $P_j \in \{P_2, P_3\}$ with $c_i(e_1 + e_2) \geq 1$ for some agent $i \in \{1, 2\}$, there exists an MMS allocation.*

PROOF. Assume w.l.o.g. that there exist two items e_1, e_2 in bundle P_3 with $c_1(e_1 + e_2) \geq 1$. Recall that for agent 1, only bundle P_1 is MMS-feasible. By Lemma 3.2, we have

$$\text{MMS}_1(M \setminus P_3, 2) \leq \text{MMS}_1(M \setminus \{e_1, e_2\}, 2) \leq \text{MMS}_1(M, 3).$$

Furthermore, we have $c_2(M \setminus P_3) < 2$. This is because agent 2 only likes P_1 which implies $c_2(P_3) > 1$. We call the *divide-and-choose*

procedure on $M \setminus P_3$ with agent 1 being the divider and agent 2 being the chooser. By Observation 3.3, it returns an allocation (X_1, X_2) such that $c_1(X_1) \leq \text{MMS}_1(M \setminus P_3, 2) \leq \text{MMS}_1(M, 3)$ and $c_2(X_2) \leq 1/2 \cdot c_2(M \setminus P_3) < 1$. Recall that $\mathbf{P}$ is an MMS partition for agent 3. Allocating bundle P_3 to agent 3 leads to an MMS allocation. $\square$

In the following, we consider the case that for any two items e_1, e_2 from the same bundle in $\{P_2, P_3\}$, we have $c_i(e_1 + e_2) < 1$ for every $i \in \{1, 2\}$.

LEMMA 3.5. *If for any two items e_1, e_2 in P_2 or P_3 , we have $c_i(e_1 + e_2) < 1$ for every $i \in \{1, 2\}$, then there exists a 9/8-AMMS allocation.*

PROOF. Recall that for agents 1 and 2, only bundle P_1 is MMS-feasible for them. We assume w.l.o.g. that $c_1(P_1) < c_1(P_2) \leq c_1(P_3)$. Note that when $c_1(P_2) \leq 9/8$, we can get a 9/8-AMMS allocation by allocating P_2 to agent 1, P_1 to agent 2 and P_3 to agent 3.

Hence, it remains to consider the case that $c_1(P_2) > 9/8$. Let $M' = P_1 \cup P_2$. We have $c_1(M') = c_1(M \setminus P_3) < 15/8$ since $c_1(P_3) \geq c_1(P_2) > 9/8$. We call the *load-balancing* procedure on items M' and cost function c_1 to get a partition $\{B_1, B_2\}$. Assume w.l.o.g. that $c_1(B_1) \geq c_1(B_2)$. Again, if $c_1(B_1) \leq 9/8$, then we are done by letting agent 2 pick the bundle she prefers in $\{B_1, B_2\}$, agent 3 receive P_3 and agent 1 receive the remaining items. In the following, we assume that $c_1(B_1) > 9/8$ and show a contradiction. We denote by e the last item that is allocated to bundle B_1 (during *load-balancing*) and let $x = c_1(e)$. We use $B'_2 \subseteq B_2$ to denote the set of items that are allocated before e . We have that $c_1(e') \geq c_1(e)$ for all $e' \in B'_2$ following the *load-balancing* procedure.

CLAIM 3.1. *We have $c_1(B'_2) \geq 2x$.*

PROOF. We consider two cases according to the size of B_1 . When $|B_1| \geq 3$, we have $c_1(B_1 - e) \geq 2x$ since there exist at least two items in B_1 that are allocated before e . Following the *load-balancing* procedure, we have $c_1(B'_2) \geq c_1(B_1 - e) \geq 2x$.

Next, we consider the case that $|B_1| < 3$. We must have $|B_1| = 2$. Otherwise we have $|B_1| = 1$ and $c_1(e) = c_1(B_1) > 9/8$, which contradicts that $c_i(e) \leq 1$. Let e' be the other item in B_1 , we have $c_1(e + e') > 9/8$. We show that $|B'_2| \geq 2$, which directly implies that $c_1(B'_2) \geq 2x$ since all items in B'_2 are allocated before e . Suppose that $|B'_2| = 1$ and e'' is the item in B'_2 (see Figure 2 for an illustration). Then we have $c_1(e'') \geq c_1(e') \geq c_1(e)$ since the algorithm allocates item e to bundle B_1 . Recall that we have $c_1(e + e') > 9/8$, which leads to that any two items in $\{e, e', e''\}$ have a total cost larger than $9/8$ for agent 1. According to the pigeon-hole principle, at least one of the bundles $\{P_1, P_2\}$ will contain two items in $\{e, e', e''\}$. Recall that we have $c_1(P_1) \leq 1$, thus bundle P_2 must contain two of items in $\{e, e', e''\}$ which contradicts the assumption that for any two items e_1, e_2 from the same bundle in P_2, P_3 , we have $c_i(e_1 + e_2) < 1$ for both $i \in \{1, 2\}$. Thus, we have $|B'_2| \geq 2$, which proves the claim. $\square$

According to the *load-balancing* procedure, we have $c_1(B'_2) \geq c_1(B_1 - e) > 9/8 - x$. By Claim 3.1, we have $c_1(B'_2) \geq 2x$. Combining the two bounds we have $3 \cdot c_1(B'_2) > \frac{18}{8}$, implying $c_1(B'_2) > 3/4$. Thus we have

$$c_1(M') = c_1(B_1) + c_1(B_2) > \frac{9}{8} + \frac{3}{4} = \frac{15}{8},$$

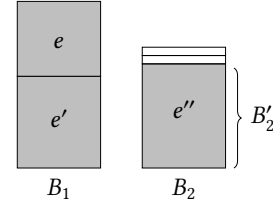


Figure 2: Illustration of B_1, B_2 and B'_2 , where the gray blocks are the three most costly items with $c_1(e'') \geq c_1(e') \geq c_1(e)$.

which contradicts that $c_1(M') = c_1(M \setminus P_3) < 15/8$. Hence, we have $c_1(B_1) \leq 9/8$ and there exists a 9/8-AMMS allocation. $\square$

Combining Lemma 3.4 and 3.5, we have proved Theorem 3.1. For completeness, we summarize our complete algorithm in Algorithm 3. We further show our analysis is tight by providing an example where Algorithm 3 returns an allocation with $c_1(B_1) = \frac{9}{8}$.

Algorithm 3: 9/8-AMMS-Three-Agents(M, N, c)

Input: An instance $\mathcal{I} = (M, N, c)$ with $|N| = 3$.
1 Let $\mathbf{P} = \{P_1, P_2, P_3\}$ be an MMS partition for agent 3;
2 **if** Agents 1 and 2 like different bundles in $\mathbf{P}$ **then**
3 Allocate agents 1 and 2 bundles that they like
 respectively;
4 Give the remaining bundle to agent 3;
5 **else**
6 Assume w.l.o.g. $c_1(P_1) \leq 1$ and $c_2(P_1) \leq 1$;
7 **if** $\exists i \in \{1, 2\}, j \in \{2, 3\}$ and $e_1, e_2 \in P_j$, such that
 $c_i(e_1 + e_2) \geq 1$ **then**
8 $X_3 \leftarrow P_j; M' \leftarrow M \setminus X_3; N' \leftarrow N \setminus \{3\};$
9 $(X_1, X_2) \leftarrow \text{Divide-and-Choose}(M', i, N' \setminus \{i\});$
10 **else**
11 $X_3 \leftarrow \arg\max_{P \in \{P_2, P_3\}} c_1(P);$
12 $M' \leftarrow M \setminus X_3; N' \leftarrow N \setminus \{3\};$
13 $\{B_1, B_2\} \leftarrow \text{Load-balancing}(M', 2, c_1);$
14 $X_2 \leftarrow \arg\min_{B \in \{B_1, B_2\}} c_2(B); X_1 \leftarrow M' \setminus X_2;$
Output: An allocation $\mathbf{X} = (X_1, X_2, X_3)$.

EXAMPLE 3.6. *Consider an instance of allocating a set of 8 items $M = \{e_k\}_{k \in [8]}$ to three agents $\{1, 2, 3\}$, where the cost functions of agents are defined as follows.*

	e_1	e_2	e_3	e_4	e_5	e_6	e_7	e_8
1	3/8	3/8	3/8	3/8	3/8	1/4	1/4	5/8
2	3/8	3/8	3/8	3/8	3/8	1/4	1/4	5/8
3	1/2	1/2	1/3	1/3	1/3	1/3	1/3	1/3

It can be verified that $\text{MMS}_i = 1$ for any $i \in N$. We have $\mathbf{P} = \{P_1, P_2, P_3\}$ with $P_1 = \{e_1, e_2\}, P_2 = \{e_3, e_4, e_5\}, P_3 = \{e_6, e_7, e_8\}$ being an MMS partition for agent 3. Note that P_1 is the only bundle that agents 1 and 2 like and $c_1(P_1) \leq c_1(P_2) \leq c_1(P_3)$ follows. Since for any two items in P_2, P_3 , the total cost is strictly less than 1 for both agents 1 and 2, we let agent 3 pick bundle P_3 and allocate

the remaining items using the load-balancing procedure. The load-balancing procedure returns bundles B_1, B_2 such that $c_i(B_1) = \frac{9}{8}$ and $c_i(B_2) = \frac{3}{4}$ for $i \in \{1, 2\}$. Since agent 2 picks first, agent 1 receives the bundle B_1 which costs $\frac{9}{8}$ for her.

4 $\frac{4}{3}$ -AMMS for Four Agents

In this section, we show the existence of $4/3$ -AMMS allocations for $n = 4$ agents. Given an instance $I = (N, M, c)$ where $N = \{1, 2, 3, 4\}$, let $P = \{P_1, P_2, P_3, P_4\}$ be an MMS partition for agent 4. We construct the MMS-feasibility graph $G = (N \cup P, E)$. If there exists a perfect matching in the graph G , then we can find an MMS allocation. Hence, in the following, we only consider the case that G does not admit a perfect matching. For ease of presentation, in the subsequent MMS-feasibility graphs in this section, we omit the edges incident to agent 4.

THEOREM 4.1. *For any instance $I = (M, N, c)$ with $n = 4$, there exists a $\frac{4}{3}$ -AMMS allocation.*

Since there is no perfect matching in the graph G , there must exist a subset $S \subseteq N$ such that $|S| > |L(S)|$, following Observation 2.6. We use S^* to denote the one with maximum size. Note that $|S^*| \neq 1$ since each agent has at least one MMS-feasible bundle in any n -partition of M . On the other hand, we have $|S^*| \neq 4$ since P is an MMS partition of agent 4. Hence to show Theorem 4.1, it suffices to consider $|S^*| \in \{2, 3\}$.

We remark that by Observation 2.7, there exists a perfect matching between $N \setminus S^*$ and $P \setminus L(S^*)$. Hence, for both cases, it remains to only consider the agents in S^* , since we can allocate each agent in $N \setminus S^*$ an MMS-feasible bundle. We show that such an operation is a valid reduction by the following observation.

OBSERVATION 4.2. *If $S^* \subseteq N$ is the maximum subset such that $|S^*| > |L(S^*)|$, then we can do a valid reduction on I by allocating each agent in $N \setminus S^*$ an MMS-feasible bundle in $P \setminus L(S^*)$.*

PROOF. By Observation 2.7, there exists a perfect matching $\mathcal{M}$ between agents $N \setminus S^*$ and bundles $P \setminus L(S^*)$. Let $I' = (M', N', c')$ be the instance after removing agents and bundles according to $\mathcal{M}$. Let $k' = |N \setminus S^*|$ and use $P_{\mathcal{M}} = \{P'_1, \dots, P'_{k'}\}$ to denote the bundles matched in $\mathcal{M}$. For any agent $i \in N'$ we have

$$c_i(M') = c_i(M) - \sum_{j \in [k']} c_i(P'_j) < n - k' = |N'|.$$

The first inequality holds because $P_{\mathcal{M}} \subseteq P \setminus L(S^*)$ and $P'_j \in P_{\mathcal{M}}$. $\square$

4.1 When $|S^*| = 2$

We first show that when $|S^*| = 2$, there exists a $\frac{4}{3}$ -AMMS allocation. Assume w.l.o.g. that $S^* = \{1, 2\}$ and $L(S^*) = \{P_1\}$ (see Figure 3). Note that following Observation 4.2, it remains to consider the reduced instance $I' = (M', N', c')$ while $N' = \{1, 2, 3\}$. We show that by calling the load-balancing procedure, we can compute an allocation such that $c_1(X_1) \leq \frac{4}{3}$ and $c_2(X_2) \leq 1$.

LEMMA 4.3. *Given a set of items M' and an agent i with $c_i(M') \leq 2$, the load-balancing procedure returns a $(\frac{4}{3}, 1)$ -partition for agent i .*

³We simply use M' to denote the remaining items after we allocated bundles to agents 3, 4. The only property of M' we will use is that $c_i(M') \leq 2$ for $i \in \{1, 2\}$, which is maintained by the valid reduction.

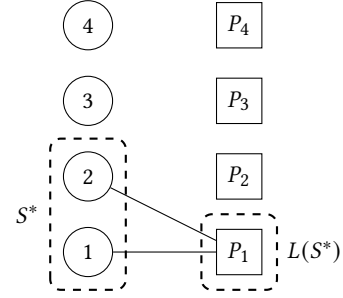


Figure 3: MMS-feasibility graph with $|S^*| = 2, |L(S^*)| = 1$.

PROOF. Let $\{B_1, B_2\}$ be the partition returned by the load-balancing procedure. We assume w.l.o.g. that $c_i(B_1) \geq c_i(B_2)$. Note that $c_i(B_2) \leq 1/2 \cdot c_i(M') \leq 1$. It remains to show that $c_i(B_1) \leq 4/3$. Suppose by contradiction that $c_i(B_1) > 4/3$. Recall that $c_i(e) \leq 1$ for all $e \in M'$, which implies that $|B_1| \geq 2$. Let e be the last item that is allocated to B_1 . According to the load-balancing procedure, we have $c_i(B_1 - e) \leq c_i(B_2) = c_i(M') - c_i(B_1) < \frac{2}{3}$.

Recall that $c_i(B_1) > 4/3$, we have $c_i(e) > 2/3$, which is a contradiction since the load-balancing procedure allocates items following the decreasing order of costs. In other words, the cost of item e cannot be larger than the cost of any item in $B_1 - e$. $\square$

LEMMA 4.4. *When $|S^*| = 2$, there exists a $\frac{4}{3}$ -AMMS allocation.*

PROOF. By Observation 4.2, it remains to consider the allocation of remaining items M' and agents 1, 2 with $c_i(M') < 2$ for $i \in \{1, 2\}$. By Lemma 4.3, calling the load-balancing procedure on c_1 returns a $(\frac{4}{3}, 1)$ -partition for agent 1. Since $c_2(M') < 2$, there exists a bundle in the partition that is MMS-feasible for agent 2. We allocate the MMS-feasible bundle to agent 2 and the remaining bundle to agent 1, which costs at most $\frac{4}{3}$ to her. $\square$

4.2 When $|S^*| = 3$

To prove Theorem 4.1, it remains to consider the case that $|S^*| = 3$, which implies that $S^* = \{1, 2, 3\}$ since all bundles in P are MMS-feasible for agent 4. We complete the proof by showing the following two lemmas.

LEMMA 4.5. *When $|S^*| = 3$ and $|L(S^*)| = 2$, there exists a $\frac{4}{3}$ -AMMS allocation.*

LEMMA 4.6. *When $|S^*| = 3$ and $|L(S^*)| = 1$, there exists a $\frac{4}{3}$ -AMMS allocation.*

Before proving the two lemma, we first show a few key observations. By Observation 4.2, it remains to consider the reduced instance $I' = (M', N', c')$ where $N' = \{1, 2, 3\}$. In the following, we propose a procedure for the reduced instance I' . We show that when I' ensures some property, there exists a $\frac{4}{3}$ -AMMS allocation for the original instance I .

LEMMA 4.7. *Given a reduced instance $I' = (M', N', c')$ with $|N'| = 3$, if there exist $T \subseteq M'$ and $i \in N'$ s.t. $c_i(T) \leq 1$ and $c_i(M' \setminus T) \leq 2$, then we can compute a $(\frac{4}{3}, 1, 1)$ -partition for agent i .*

PROOF. We call the load-balancing procedure on items $M' \setminus T$ with respect to c_i . By Lemma 4.3, the returned partition $\{B_1, B_2\}$

is a $(\frac{4}{3}, 1)$ -partition for agent i . Hence for agent i , the partition $\{T, B_1, B_2\}$ is a $(\frac{4}{3}, 1, 1)$ -partition. $\square$

LEMMA 4.8. *Given a reduced instance $I' = (M', N', c')$ with $|N'| = 3$, if there exists a $(\frac{4}{3}, 1, 1)$ -partition for some agent $i \in N'$, then we can compute a $\frac{4}{3}$ -AMMS allocation for the original instance $I = (M, N, c)$.*

PROOF. We remark that agent 4 receives a bundle that is MMS-feasible only for her, hence it remains to consider the reduced instance I' . Let $\{B'_1, B'_2, B'_3\}$ be a $(\frac{4}{3}, 1, 1)$ -partition (of items M') for agent 1. We consider which bundles agents 2 and 3 like. If there exists a perfect matching between agents $\{2, 3\}$ and bundles $\{B'_1, B'_2, B'_3\}$, we allocate agents 2 and 3 each a bundle following the matching and agent 1 receives the remaining bundle. The bundle agent 1 received costs at most $4/3$ to her. Consider otherwise that there is only one bundle being MMS-feasible for agents 2 and 3. There must exist a bundle that is MMS-feasible for agent 1 but not for agents 2 and 3. After assigning such a bundle to agent 1, the total cost of the remaining items is less than 2 for agents 2 and 3. We call the *load-balancing* procedure on the remaining items w.r.t. c_2 . By Lemma 4.3, the returned partition is a $(\frac{4}{3}, 1)$ -partition for agent 2. By letting agent 3 pick the bundle she prefers, agent 2 receives the remaining bundle, which costs at most $4/3$. $\square$

We summarize the above procedure in Algorithm 4.

Algorithm 4: $4/3$ -AMMS-Three-Agents($I', \{B'_1, B'_2, B'_3\}$)

Input: Reduced instance $I' = (M', N', c')$, a partition $\{B'_1, B'_2, B'_3\}$ that is a $(4/3, 1, 1)$ -partition for agent 1.

- 1 **if** There exists a perfect matching between agents 2, 3 and bundles $\{B'_1, B'_2, B'_3\}$ **then**
- 2 Allocate the bundle to agents 1 and 2, respectively,
 based on the perfect matching;
- 3 Agent 1 receives the remaining one;
- 4 **else**
- 5 Let agent 1 pick the bundle that she likes while agents 2
 and 3 dislike;
- 6 $M' \leftarrow M' \setminus X_1$;
- 7 $(X_2, X_3) \leftarrow \text{Load-Balancing}(M', 2, c_2)$;

Output: (X_1, X_2, X_3)

Given Lemma 4.7 and 4.8, in the following part of this section, we focus on computing $(\frac{4}{3}, 1, 1)$ -partition for some agents. We first consider the case where $|S^*| = 3$ and $|L(S^*)| = 2$, in which there are two bundles, say P_3 and P_4 , that are MMS-feasible only for agent 4.

LEMMA 4.5. *When $|S^*| = 3$ and $|L(S^*)| = 2$, there exists a $\frac{4}{3}$ -AMMS allocation.*

PROOF. We first consider the case that each agent in S^* only has one MMS-feasible bundle. Since $|S^*| = 3$, $|L(S^*)| = 2$, there must exist two agents that like the same bundle, say that agents 1, 2 like bundle P_1 . As Figure 4a shows, we can allocate bundle P_2 to agent 3 and bundle P_4 to agent 4, which guarantees MMS for both of them. By Observation 4.2, it remains to consider the allocation of items $P_1 \cup P_3$ to agents 1, 2 with $c_i(P_1 \cup P_3) < 2$ for both $i \in \{1, 2\}$. By

Lemma 4.3, we can find a $(\frac{4}{3}, 1)$ -partition on items $P_1 \cup P_3$ for agents 1. Letting agent 2 pick first and agent 1 receive the remaining one results in a $4/3$ -AMMS allocation.

Next, we move to the case where there exists an agent that likes two bundles (see Figure 4b). We assume w.l.o.g. that agent 1 likes two bundles P_1, P_2 and $c_1(P_1) \leq c_1(P_2) \leq c_1(P_3) \leq c_1(P_4)$. By allocating bundle P_4 to agent 4, it remains to consider $I' = (M', N', c')$ with $N' = \{1, 2, 3\}$, $M' = M \setminus P_4$ since $c_i(M \setminus P_4) < 3$ for all $i \in \{1, 2, 3\}$. Notice that we have $c_1(P_1) + c_1(P_3) \leq c_1(P_2) + c_1(P_4)$, which implies that $c_1(P_1) + c_1(P_3) \leq \frac{1}{2} \cdot c_1(M) \leq 2$. Combined with the assumption that $c_1(P_2) < 1$, by Lemma 4.7, we can compute a $(\frac{4}{3}, 1, 1)$ -partition for agent 1. Based on the partition, we can find a $4/3$ -AMMS allocation by Lemma 4.8. $\square$

It remains to consider the case that all agents 1, 2, 3 like the same bundle, say P_1 (See Figure 4c). We remark that this case is the most complicated and needs a more careful analysis. One of the differences compared with the previous analysis is that we not only need the valid reduction property, i.e., $c_i(M') \leq 3$ for all $i \in N'$, but also need to specify which bundle is allocated to agent 4. To specify such a bundle, we use the *atomic bundle* framework introduced by Feige and Norkin [23]. More specifically, the *atomic bundle* is the intersection of two bundles, each of them from a different agent's MMS partitions. Note that in this case, agents 1, 2, 3 are symmetric. In the following, we fix agent 1 and let $\mathbf{B} = \{B_1, B_2, B_3, B_4\}$ be an MMS partition for agent 1. We take the intersection of bundles in $\mathbf{B}$ and bundles in $\mathbf{P}$, i.e., $b_{ij} = B_i \cap P_j$ for all $i, j \in \{1, 2, 3, 4\}$. We show the atomic bundles as a matrix in Figure 5a.

LEMMA 4.6. *When $|S^*| = 3$ and $|L(S^*)| = 1$, there exists a $\frac{4}{3}$ -AMMS allocation.*

PROOF. We first consider the case that there exists an atomic bundle b_{ij} , such that $c_1(P_j \setminus b_{ij}) > 1$ for some $i \in \{1, 2, 3, 4\}$, $j \in \{2, 3, 4\}$. We show that in this case, Lemma 4.7 can be applied to find a $4/3$ -AMMS allocation. Note that b_{ij} appears in P_2, P_3, P_4 symmetrically. We assume w.l.o.g. that $b_{42} \subseteq P_2$ is such an atomic bundle (see Figure 5b). By assigning P_2 to agent 4, it remains to consider the reduced instance $I' = (M', N', c')$ where $M' = M \setminus P_2$ and $N' = \{1, 2, 3\}$. Let $T = b_{41} \cup b_{43} \cup b_{44}$, we have $c_1(T) \leq 1$ since $T \subseteq B_4$ while $c_1(B_4) \leq 1$. Moreover, since $c_1(P_2 \setminus b_{42}) > 1$, we have

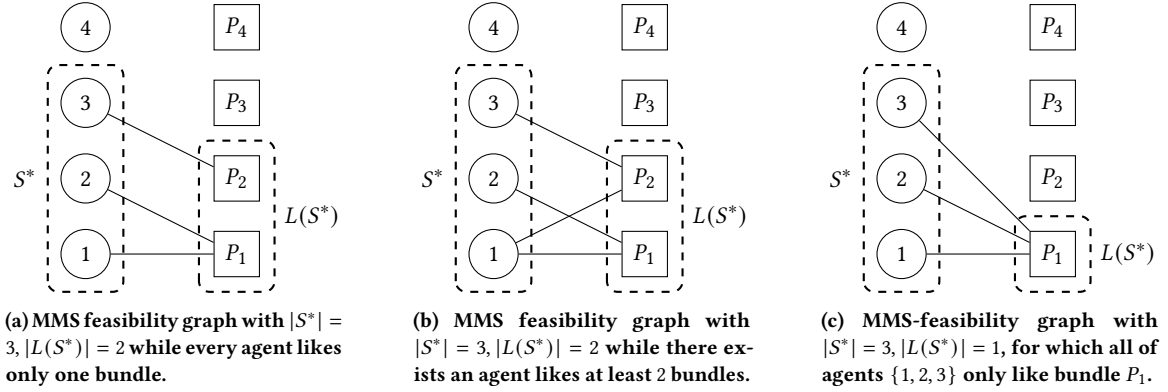
$$c_1(M' \setminus T) = c_1(B_1 \cup B_2 \cup B_3 \setminus P_2) < 2.$$

See Figure 5c for an illustration. By applying Lemma 4.7, we can compute a $(\frac{4}{3}, 1, 1)$ -partition for agent 1. Based on the partition, we can find a $4/3$ -AMMS allocation by Lemma 4.8.

It remains to consider that for any b_{ij} where $i \in \{1, 2, 3, 4\}$, $j \in \{2, 3, 4\}$, we have $c_1(P_j \setminus b_{ij}) \leq 1$. We assume w.l.o.g. that $c_1(P_1) \leq c_1(P_2) \leq c_1(P_3) \leq c_1(P_4)$. By assigning P_4 to agent 4, it remains to consider the reduced instance $I' = (M', N', c')$ where $M' = M \setminus P_4$ and $N' = \{1, 2, 3\}$. Again, we show that there exists a $(\frac{4}{3}, 1, 1)$ -partition for agent 1. Let $c_1(P_1) = \beta$. We have

$$c_1(P_2) + c_1(P_3) = c_1(M) - c_1(P_1) - c_1(P_4) \leq \frac{2(c_1(M) - \beta)}{3} \leq \frac{2(4 - \beta)}{3},$$

where the first inequality holds because $c_1(P_1) + c_1(P_4) \geq \frac{c_1(M) + 2\beta}{3}$ and the second inequality holds because $c_1(M) \leq 4$. Let b_2 and

Figure 4: MMS feasibility graphs with $|S^*| = 3, |L(S^*)| \leq 2$.

	≤ 1	> 1	> 1	> 1
	P_1	P_2	P_3	P_4
B_1	b_{11}	b_{12}	b_{13}	b_{14}
B_2	b_{21}	b_{22}	b_{23}	b_{24}
B_3	b_{31}	b_{32}	b_{33}	b_{34}
B_4	b_{41}	b_{42}	b_{43}	b_{44}

(a) The atomic bundles presented as a matrix.

	P_1	P_2	P_3	P_4
B_1	b_{11}	b_{12}	b_{13}	b_{14}
B_2	b_{21}	b_{22}	b_{23}	b_{24}
B_3	b_{31}	b_{32}	b_{33}	b_{34}
B_4	b_{41}	b_{42}	b_{43}	b_{44}

(b) The gray bundles cost at most 3 to agent 1.

	P_1	P_2	P_3	P_4
B_1	b_{11}	b_{12}	b_{13}	b_{14}
B_2	b_{21}	b_{22}	b_{23}	b_{24}
B_3	b_{31}	b_{32}	b_{33}	b_{34}
B_4	b_{41}	b_{42}	b_{43}	b_{44}

(c) The gray bundles cost at most 2 to agent 1

Figure 5: Each atomic bundle is an intersection of a bundle in B (an MMS partition of agent 1) and a bundle in P (an MMS partition of agent 4). We remark that P_1 is the bundle that agent 1 likes while she dislikes all other bundles P_2, P_3, P_4 .

b_3 be the atomic bundle with the minimum cost in P_2 and P_3 respectively, i.e., $b_2 = \operatorname{argmin}\{c_1(b) : b \in \{b_{12}, b_{22}, b_{32}, b_{42}\}\}$, and $b_3 = \operatorname{argmin}\{c_1(b) : b \in \{b_{13}, b_{23}, b_{33}, b_{43}\}\}$.

- Suppose $\beta \leq \frac{4}{5}$. We have

$$c_1(b_2) + c_1(b_3) \leq \frac{1}{4} \cdot (c_1(P_2) + c_1(P_3)) \leq \frac{1}{4} \cdot \frac{2(4-\beta)}{3} = \frac{4-\beta}{6},$$

which implies that $c_1(P_1) + c_1(b_2) + c_1(b_3) \leq \beta + \frac{4-\beta}{6} \leq \frac{4}{3}$. Combine with $c_1(P_2 \setminus b_2) \leq 1$ and $c_1(P_3 \setminus b_3) \leq 1$, $\{P_1 \cup b_2 \cup b_3, P_2 \setminus b_2, P_3 \setminus b_3\}$ is a $(\frac{4}{3}, 1, 1)$ -partition for agent 1.

- Suppose $\beta > \frac{4}{5}$. We have

$$c_1(b_3) \leq \frac{1}{4} \cdot c_1(P_3) \leq \frac{1}{4} \cdot \frac{4-\beta - c_1(P_2)}{2}.$$

Note that $c_1(P_2) \leq c_1(P_3)$ while $c_1(P_2) + c_1(P_3) \leq \frac{2(4-\beta)}{3}$.

Hence we have $c_1(P_2) \leq \frac{4-\beta}{3}$ and

$$c_1(P_2) + c_1(b_3) \leq \frac{4-\beta + 7c_1(P_2)}{8} \leq \frac{5}{3} - \frac{5}{12}\beta \leq \frac{4}{3}.$$

Recall that $c_1(P_3 \setminus b_3) \leq 1$ and $c_1(P_1) < 1$. Hence $\{P_1, P_2 \cup b_3, P_3 \setminus b_3\}$ is a $(\frac{4}{3}, 1, 1)$ -partition for agent 1. $\square$

5 Conclusion

In this paper, we study the all-but-one MMS (AMMS) allocations where we guarantee all but one agent to achieve their maximin share. We show that the approximation ratio is at most $9/8$ for three agents, $4/3$ for four agents and $\frac{(n+1)^2}{4n}$ for $n \geq 5$ agents. The AMMS framework provides a novel approach to examining the approximation of MMS allocations from an optimization perspective. This alternative perspective may further deepen our understanding of the MMS concept. Moreover, as a stronger notion, AMMS naturally inherits all existing counterexamples of MMS. Given that the existing hard instances admit allocations that are very close to being MMS, we conjecture that $O(1)$ -AMMS allocations always exist for a general number of agents. We also believe that the AMMS framework may inspire further research on other MMS relaxations, such as MMS allocations with item disposal or MMS allocations with subsidies. It is noteworthy that any AMMS allocation can be converted into an MMS allocation with unallocated items (or with subsidies), only requiring addressing the agent who is not guaranteed her MMS. Another interesting direction is to consider balanced relaxations to avoid overburdening a single agent, e.g., guaranteeing $O(1)$ -MMS for k agents.

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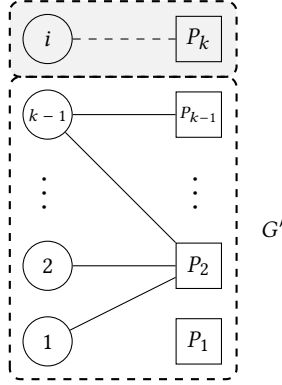
A $\frac{(n+1)^2}{4n}$ -AMMS for $n \geq 5$ Agents

In this section, we consider the instances with a general number of agents. Let $\gamma = \frac{(n+1)^2}{4n}$. Given any instance $I = (M, N, c)$, we show that γ -AMMS allocations are guaranteed to exist, by proposing Algorithm 5. The algorithm runs in iterations, while during each iteration it either returns a γ -AMMS allocation (and terminates), or reduces the input instance (a reduced instance) to a new reduced instance with a smaller size. We remark that the original instance $I = (M, N, c)$ can also be viewed as a reduced instance since $c_i(M) \leq n = |N|$ for all $i \in N$. Before introducing the main result, we first give an overview of the algorithm for general n .

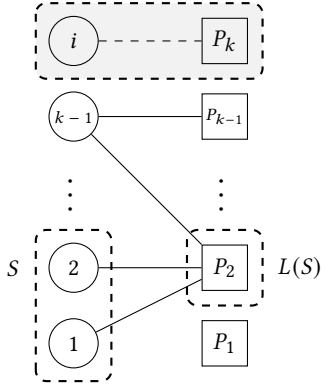
We remark that if there exists a perfect matching in G' , then we can directly find a γ -AMMS allocation, by allocating P_k to agent i and the others following the perfect matching. If there is no perfect matching, we claim that we can make a valid reduction on I' , which means that we can remove at least one agent and one bundle of items from I' . The algorithm terminates after at most n rounds.

THEOREM A.1. *For any instance with $n \geq 5$ agents, there exist γ -AMMS allocations.*

To prove Theorem A.1, we first show that the partition $P = \{P_1, \dots, P_k\}$ we output during each iteration, is a $(\gamma, 1, \dots, 1)$ -partition for some agent $i \in N'$. We remark that at the beginning of any iteration, we have a reduced instance $I' = (M', N', c')$ as input.



(a) The MMS-feasibility graph G' is a subgraph of G that excludes agent i , bundle P_k and all their incident edges, e.g., $G' = (N \setminus \{i\} \cup P \setminus \{P_k\}, E)$. The dashed edge between i and P_k in the figure denotes that $c_i(P_k) \leq \gamma$.



(b) An illustration for S and $L(S)$. We can let agents in $N' \setminus \{1, 2, i\}$ pick bundles following the perfect matching $\mathcal{M}$ and agent i pick one from the remaining bundles in $P' \setminus L(S)$. It remains to consider a reduced instance with only agents $\{1, 2\}$.

Figure 6: Illustrations for subgraph G' and S with $L(S)$.

Algorithm 5: γ -AMMS- n -Agents(M, N, c)

Input: An original instance $\mathcal{I} = (M, N, c)$

- 1 $M' \leftarrow M; N' \leftarrow N; c' \leftarrow c;$
- 2 Let $\mathcal{I}' \leftarrow (M', N', c')$ and $k \leftarrow |N'|;$
- 3 **while** $N' \neq \emptyset$ **do**
- 4 Pick an arbitrary agent $i \in N';$
- 5 Find a $(\gamma, 1, \dots, 1)$ -partition P for agent $i;$
- 6 Construct an MMS-feasibility graph G' between $N' \setminus \{i\}$ and $P \setminus \{P_k\};$
- 7 **if** there exists a perfect matching $\mathcal{M}$ in G' **then**
- 8 Allocate bundles to the corresponding agents in $\mathcal{M}$ and allocate P_k to agent $i;$
- 9 **return** $(X_1, \dots, X_n);$
- 10 $(\hat{M}, \hat{N}, \hat{c}), \{X_i\}_{i \in N' \setminus \hat{N}} \leftarrow \text{Valid-Reduction}(G, i);$
- 11 // Algorithm 6
- 12 Update $M' \leftarrow \hat{M}, N' \leftarrow \hat{N}, c' \leftarrow \hat{c};$
- 13 Let $\mathcal{I}' \leftarrow (M', N', c')$ and $k \leftarrow |N'|;$

Output: An allocation $\mathbf{X} = (X_1, \dots, X_n)$

LEMMA A.2. Given a reduced instance $\mathcal{I}' = (M', N', c')$, we can compute a partition $\mathbf{P} = \{P_1, \dots, P_k\}$ that is a $(\gamma, 1, \dots, 1)$ -partition for some agent $i \in N'$.

For continuity of presentation, we defer the proof of Lemma A.2 to the end of this section. Given such a $(\gamma, 1, \dots, 1)$ -partition $\mathbf{P} = \{P_1, \dots, P_k\}$, we construct an MMS-feasibility graph G . We mainly focus on the subgraph G' with agents $N' \setminus \{i\}$, the set of bundles $\mathbf{P} \setminus \{P_k\}$, and the edges between them (see Figure 6a). If there exists a perfect matching $\mathcal{M}$ in the graph G' , by allocating P_k to agent i and the other bundles following the matching, we have a γ -AMMS allocation. It remains to consider that there is no perfect matching in G' , in which case we run Algorithm 6 to reduce the instance.

Algorithm 6: Valid-Reduction(G', i)

Input: An MMS-feasibility graph $G' = ((N' \setminus \{i\}) \cup (P \setminus \{P_k\}), E)$ and agent i

- 1 Let $\hat{N} = N' \setminus \{i\}$ and $\hat{P} = P \setminus \{P_k\};$
- 2 **if** there is no perfect matching between $\hat{N}$ and $\hat{P}$ **then**
- 3 Let $S = \text{argmax}_{\{S' \subseteq N' \setminus \{i\} : |S'| > |L(S')|\}} |S'|;$
- 4 Find a perfect matching $\mathcal{M}$ between $(\hat{N} \setminus \{i\}) \setminus S$ and $\hat{P} \setminus L(S);$
- 5 Let $N_{\mathcal{M}}, P_{\mathcal{M}}$ be the set of agents and bundles in $\mathcal{M};$
- 6 Allocate bundles $P_{\mathcal{M}}$ to agents in $N_{\mathcal{M}}$ following $\mathcal{M};$
- 7 $\hat{M} \leftarrow M' \setminus P_{\mathcal{M}}, \hat{N} \leftarrow N' \setminus N_{\mathcal{M}};$
- 8 **else**
- 9 There exists a γ -AMMS allocation;

Output: An instance $\hat{\mathcal{I}} = (\hat{M}, \hat{N}, \hat{c})$ and $\{X_i\}_{i \in N' \setminus \hat{N}}$

LEMMA A.3. If there is no perfect matching in the graph G' , Algorithm 6 can reduce the reduced instance $\mathcal{I}'$ to a new reduced instance $\hat{\mathcal{I}}$ with a strictly smaller size.

PROOF. Let $\hat{N} = N' \setminus \{i\}$ and $\hat{P} = P \setminus \{P_k\}$. If there exists no perfect matching between $\hat{N}$ and $\hat{P}$, by Hall's theorem, we can find a maximum subset $S \subseteq \hat{N}$ s.t. $|S| > |L(S)|$. Hence we must have $|L(S)| \leq |S| - 1$ and $|\hat{P} \setminus L(S)| = |\hat{N}| - |L(S)| \geq |\hat{N}| - |S| + 1$. By Observation 2.7, there exists a perfect matching between agents $\hat{N} \setminus S$ and bundles $\hat{P} \setminus L(S)$. Recall that all bundles in $\hat{P} \setminus L(S)$ are MMS-feasible for agent i , hence letting her pick any one of the remaining bundles in $\hat{P} \setminus L(S)$ results in a perfect matching $\mathcal{M}$.

On the other hand, the matching size is $|\hat{N}| - |S| + 1 \geq 1$, which implies that the returned instance $\hat{\mathcal{I}}$ has a strictly smaller size. To prove Lemma A.3, it suffices to show the reduction in Algorithm 6 is valid. Since the bundles allocated in Algorithm 6 are not MMS-feasible for agents in S , the proportional share of agents in S will not increase. On the other hand, any agent in $N' \cup \{i\} \setminus S$ receives a bundle in Algorithm 6 while the bundle costs at most 1 for her. Hence, the reduction in Algorithm 6 is a valid reduction. $\square$

Proof of Theorem A.1: Following Lemma A.2, given any reduced instance $\mathcal{I}'$ with $|N'| = k$, for any agent $i \in N'$, there exists a $(\gamma, 1, \dots, 1)$ -partition of size k for agent i .

According to Lemma A.3, we can either get a γ -AMMS allocation on instance $\mathcal{I}'$ or run Algorithm 6 to get a smaller reduced instance $\mathcal{I}'$. If we get a γ -AMMS allocation, we are done. If we run Algorithm 6, we can get a smaller instance $\mathcal{I}'$ by removing at least one pair of agent and bundle. These two cases directly imply that the Algorithm 5 will terminate after at most n rounds. According to the reduction rules in Algorithm 6, every agent excluded from the reduced instance gets an MMS-feasible bundle. Thus, we can conclude that the returned allocation of Algorithm 5 is a γ -AMMS allocation for a general number of agents. ■

It remains to prove Lemma A.2. Based on the size of the reduced instance $|N'|$, we introduce different algorithms to compute a $(\gamma, 1, \dots, 1)$ -partition for the chosen agent i . The first algorithm is called *Partition-Merging* (see Algorithm 7), in which we involve the MMS partition $\mathbf{P}$ of agent i for the original instance $\mathcal{I}$. We consider the “current state” of the original MMS partition, i.e., $P'_j = P_j \cap M'$ for all $j \in N$. We merge $n - k + 1$ bundles with the smallest cost to be one, yielding a k -partition of items M' . We show that when $|N'| > \lfloor \frac{n+1}{2} \rfloor$, it gives a $(\gamma, 1, \dots, 1)$ -partition for agent i .

Algorithm 7: Partition-Merging($\mathcal{I}, \mathcal{I}', i$)

Input: Original instance $\mathcal{I} = (M, N, c)$, reduced instance $\mathcal{I}' = (M', N', c')$ and agent i

- 1 Let $k \leftarrow |N'|$;
 - 2 Let $\mathbf{P} = \{P_1, \dots, P_n\}$ be an MMS partition of agent i for instance $\mathcal{I}$;
 - 3 Set $P'_j = P_j \cap M'$ for all $j \in N$;
 - 4 Sort and rename bundles in $\{P'_j\}_{j \in N}$ such that $c_i(P'_1) \geq c_i(P'_2) \geq \dots \geq c_i(P'_n)$;
- Output:** A partition $\{P'_1, \dots, P'_{k-1}, \bigcup_{j=k}^n P'_j\}$
-

LEMMA A.4. *Given original instance $\mathcal{I} = (M, N, c)$ and a reduced instance $\mathcal{I}' = (M', N', c')$ with $|N'| > \lfloor \frac{n+1}{2} \rfloor$, Algorithm 7 returns $\mathbf{P} = \{P_1, \dots, P_k\}$ that is a $(\gamma, 1, \dots, 1)$ -partition for agent i .*

PROOF. Let $\mathbf{P} = \{P_1, \dots, P_n\}$ be an MMS partition of M for agent i . Based on $\mathbf{P}$, we construct a partition $\mathbf{P}' = \{P'_1, \dots, P'_n\}$ of items M' where $P'_j = P_j \cap M'$ for all $j \in [n]$. We assume w.l.o.g. that $\mathbf{P}'$ is sorted in non-increasing order of $c_i(P'_j)$, i.e., $c_i(P'_1) \geq \dots \geq c_i(P'_n)$.

Let $k = |N'|$. By merging the last $n - k + 1$ bundles in $\mathbf{P}'$, we have $\{P'_1, \dots, P'_{k-1}, \bigcup_{j=k}^n P'_j\}$ being a k -partition of M' . Clearly, we have $c_i(P'_j) \leq 1$ for every $j \in [k - 1]$ since $P'_j \subseteq P_j$ and $c_i(P_j) \leq 1$ for all $j \in [k]$. For the last merged bundle $\bigcup_{j=k}^n P'_j$, we have

$$c_i\left(\bigcup_{j=k}^n P'_j\right) \leq \frac{n-k+1}{n} \cdot \sum_{j=1}^n c_i(P'_j) < \frac{n-k+1}{n} \cdot k = \frac{-k^2+(1+n)k}{n},$$

where the first inequality holds because $\mathbf{P}'$ is sorted in non-increasing order of $c_i(P'_j)$ and the second inequality holds because $\sum_{j=1}^n c_i(P'_j) = c_i(M') < k$. Recall that $k > \lfloor \frac{n+1}{2} \rfloor$, we have

$$c_i\left(\bigcup_{j=k}^n P'_j\right) < \frac{-k^2+(1+n)k}{n} < \frac{(n+1)^2}{4n} = \gamma,$$

which completes our proof. ■

In the following, we consider the case that $|N'| \leq \lfloor \frac{n+1}{2} \rfloor$, for which we propose the *Capped-Bag-Filling* (see Algorithm 8) to compute a $(\gamma, 1, \dots, 1)$ -partition for some agent i . At the beginning of the algorithm, we initialize $|N'|$ empty bundles. We allocate items from the largest cost to the minimum, w.r.t. to c_i , while we only assign item e to bundle P if $c_i(P + e) \leq 1$. When it is not possible to allocate any item to any bundle, the remaining items will be assigned to the bundle with the lowest cost.

Algorithm 8: Capped-Bag-Filling($\mathcal{I}', i$)

Input: A reduced instance $\mathcal{I}' = (M', N', c')$ and an agent i

- 1 Let $k \leftarrow |N'|$, Initialize $\mathbf{P} = \{P_j\}_{j \in [k]}$ with $P_j \leftarrow \emptyset$;
 - 2 Sort items with $c_i(e_1) \geq \dots \geq c_i(e_{|M'|})$;
 - 3 **for** $j' = 1$ **to** $|M'|$ **do**
 - 4 **if** there exists a bundle P_j such that $c_i(P_j + e_{j'}) \leq 1$ **then**
 - 5 $P_j \leftarrow P_j + e_{j'}$; $M' \leftarrow M' - e_{j'}$;
 - 6 Sort $\mathbf{P}$ such that $c_i(P_1) \geq \dots \geq c_i(P_k)$;
 - 7 $P_k \leftarrow P_k \cup M'$;
- Output:** A partition $\{P_1, \dots, P_k\}$
-

LEMMA A.5. *Given a reduced instance $\mathcal{I}' = (M', N', c')$ with $|N'| \leq \lfloor \frac{n+1}{2} \rfloor$, Alg. 8 returns $\mathbf{P} = \{P_1, \dots, P_k\}$ that is a $(\gamma, 1, \dots, 1)$ -partition for agent i .*

PROOF. By Algorithm 8, given an instance $\mathcal{I}' = (M', N', c')$ and an arbitrary agent $i \in N'$ as input, we get a k -partition $\{P_1, \dots, P_k\}$ of items M' . We denote by P'_k the remaining items after the Bag-Filling phase. By Algorithm 8, P'_k is assigned to bundle P_k . For any item $e \in P'_k$, we have $c_i(P_j + e) > 1$ for all $j \in [k - 1]$ and $c_i(P_k \setminus P'_k) + c_i(e) > 1$ following the algorithm.

CLAIM A.1. $|P'_k| \leq k - 1$.

PROOF. Suppose by contradiction that $|P'_k| \geq k$. Let $\{e_1, \dots, e_k\}$ be a subset of bundle P'_k . By allocating item e_i to bundle P_i for all $i \in [k - 1]$ and e_k to bundle $P_k \setminus P'_k$, we have $k < \sum_{j \in [k-1]} c_i(P_j + e_j) + c_i(P_k \setminus P'_k + e_k) \leq c_i(M')$ which contradicts the fact that $\mathcal{I}'$ is a reduced instance. ■

For every $j \in [k]$, let e'_j be the item with the largest cost in bundle P_j . Let $P = \bigcup_{j \in [k]} e'_j$ and $\hat{M} = P \cup P'_k$. For any $e' \in P$ and any $e \in P'_k$, we have $c_i(e') \geq c_i(e)$ since the items are allocated in descending order of their costs. Then we have

$$\begin{aligned} c_i(P'_k) &\leq \frac{|P'_k|}{|M|} \cdot c_i(\hat{M}) = \frac{|P'_k|}{|P'_k| + k} \cdot c_i(\hat{M}) = \frac{1}{1 + \frac{k}{|P'_k|}} \cdot c_i(\hat{M}) \\ &\leq \frac{k-1}{2k-1} \cdot c_i(\hat{M}) \leq \frac{k-1}{2k-1} \cdot k, \end{aligned} \quad (1)$$

where the second inequality holds due to Claim A.1 and the last inequality holds due to $\hat{M} \subseteq M'$ and $c_i(M') \leq k$. Since P'_k was allocated to the bundle with the lowest cost after the Bag-Filling phase, we have $c_i(P_k \setminus P'_k) \leq c_i(P_j)$ for every $j \in [k - 1]$, which implies that

$$c_i(P_k \setminus P'_k) \leq \frac{c_i(M') - c_i(P'_k)}{k} \leq \frac{k - c_i(P'_k)}{k}. \quad (2)$$

Then we have

$$\begin{aligned} c_i(P_k) &= c_i(P'_k) + c_i(P_k \setminus P'_k) \leq \frac{k + (k-1) \cdot c_i(P'_k)}{k} \\ &\leq \frac{k + (k-1) \cdot \frac{k-1}{2k-1} \cdot k}{k} = \frac{k^2}{2k-1} \end{aligned}$$

where the first inequality holds due to Equation (2) and the second inequality holds due to Equation (1). Since $k \leq \lfloor \frac{n+1}{2} \rfloor$, we have

$$c_i(P_k) \leq \frac{k^2}{2k-1} \leq \frac{(n+1)^2}{4n} = \gamma,$$

which completes the proof. $\square$

B Extension to Allocation of Goods

In this section, we discuss both positive and negative results regarding the extension to the allocation of goods. Suppose each agent i has a valuation function $v_i : 2^M \rightarrow \mathbb{R}^+ \cup \{0\}$ on the sets of goods, the MMS value of agent i is defined as follows.

DEFINITION B.1. For an instance $\mathcal{I} = (M, N, \mathbf{v})$ with n agents, the maximin share of agent i with valuation function v_i is defined as

$$\text{MMS}_i(M, n) = \max_{P \in \Pi_n(M)} \min_{j \in [n]} \{v_i(P_j)\}.$$

When M and n are clear, we write it as MMS_i . Likewise, we assume w.l.o.g. that $\text{MMS}_i = 1$, thus $v_i(M) \geq n$ for all $i \in N$. We next introduce the definition of α -AMMS for goods.

DEFINITION B.2 (α -AMMS FOR GOODS). An allocation $\mathbf{X}$ is α -approximate all-but-one maximin share (α -AMMS), if there exists an agent $i \in N$ with $v_i(X_i) \geq \alpha$ and $v_j(X_j) \geq 1$ holds $\forall j \in N \setminus \{i\}$.

Recall that under the notion of (α, β) -MMS [34], α fraction of the agents are guaranteed a β approximation of their MMS. Therefore, α -AMMS (for any $\alpha \in (0, 1)$) is a stronger notation than $(\frac{n-1}{n}, 1)$ -MMS because we further put a guarantee on the utility of the unsatisfied agent. The existence of $(\frac{n-1}{n}, 1)$ -MMS allocations remains a major open problem. In this section, we extend part of our results for chores to the allocations of goods.

THEOREM B.3. For the allocation of goods to three agents, there exist $6/7$ -AMMS allocations.

We first argue that it suffices to consider the identical ordering (IDO) instances only.

DEFINITION B.4 (IDO INSTANCES). An instance is called IDO if all agents have the same ordering of preference over all items, i.e., $v_i(e_1) \geq v_i(e_2) \geq \dots \geq v_i(e_m)$, $\forall i \in [3]$.

It was shown by Bouveret *et al.* [17] that if there exists an α -MMS allocation for IDO instances, then there exists an α -MMS allocation for general instances. Following exactly the same proof, it suffices to consider IDO instances only.

LEMMA B.5. If there exist α -AMMS allocations for IDO instances, then there exist α -AMMS allocations for general instances.

We can further assume that the instance is *normalized*, where for all agents, we have $\text{MMS}_i = 1$ and $v_i(M) = 3$, as shown in [3, 4]. Note that the assumption implies that for every agent i , every bundle in her MMS partition has value exactly 1, and $v_i(e) \leq 1$ for all $e \in M$. Moreover, we can assume w.l.o.g. that $v_i(e_3 + e_4) \leq 1$ for all agents $i \in [3]$. By the pigeon-hole principle, in the MMS partition

of every agent i , there must exist a bundle P containing two items from $\{e_1, e_2, e_3, e_4\}$. Since $v_i(P) = 1$, we have $v_i(e_3 + e_4) \leq 1$.

Now we proceed to present our algorithm. We first find an MMS partition $\mathbf{P} = \{P_1, P_2, P_3\}$ for agent 3 and construct the MMS-feasibility graph G based on $\mathbf{P}$. By similar arguments from Section 3, it suffices to consider the case that all agents like bundle P_1 while bundles P_2 and P_3 are MMS-feasible only to agent 3. We assume w.l.o.g. that $v_1(P_1) \geq v_1(P_2) \geq v_1(P_3)$. We can further assume that $v_1(P_3) < 6/7$, as otherwise we can get a $6/7$ -AMMS allocation by allocating P_3 to agent 1, P_1 to agent 2, and P_2 to agent 3. Let $M' = P_1 \cup P_2$. We have $v_1(M') = v_1(M) - v_1(P_3) > 15/7$. We call the *load-balancing* procedure on M' and v_1 to get a partition $\{B_1, B_2\}$ such that $v_1(B_1) \geq v_1(B_2)$. Again, if $v_1(B_2) \geq 6/7$, we can assign bundle P_3 to agent 3, let agent 2 pick a bundle between $\{B_1, B_2\}$, and agent 1 gets the other bundle, which results in a $6/7$ -AMMS allocation. Similarly, if $v_2(B_2) \geq 6/7$, we can assign P_3 to agent 3, B_1 to agent 1 and B_2 to agent 2. Now suppose that $v_1(B_2) < 6/7$, we have $v_1(B_1) = v_1(M') - v_1(B_2) > 9/7$. Next, we show a few key observations for the bundles $\{B_1, B_2\}$.

Let e be the last item that is allocated to bundle B_1 . Let $B'_2 \subseteq B_2$ be the set of items that are allocated before e . We have $v_1(B'_2) \geq v_1(B_1 - e) > 9/7 - v_1(e)$, which implies $v_1(e) > 3/7$. Therefore, all items allocated before e have a value larger than $3/7$. Since $v_1(B_2) < 6/7$, we conclude that $|B'_2| = 1$ (say, $B'_2 = \{e'_1\}$) and there is only one item (say, e'_2) allocated to B_1 before e . For consistency, we rename e as e'_3 . Note that e'_1, e'_2, e'_3 are the three most valuable items in M' , and we have $v_1(e'_1) \geq v_1(e'_2) \geq v_1(e'_3)$. Recall that the instance is IDO, e.g., all agents consider e'_1, e'_2, e'_3 as the three most valuable items in M' . In the following, we show more properties.

CLAIM B.1. Item e'_1 is the most valuable item in M , i.e., $e'_1 = e_1$.

PROOF. Suppose otherwise, we have $e_1 \in P_3$ (recall that $M' = P_1 \cup P_2$). Then we have $v_1(B_1) = v_1(e'_2 + e'_3) \leq v_1(e_3 + e_4) < 1$ which contradicts $v_1(B_1) > 9/7$. $\square$

CLAIM B.2. We have $v_3(B_2) \geq 1$.

PROOF. By the pigeon-hole principle, one of P_1 and P_2 must contain two items from $\{e'_1, e'_2, e'_3\}$. Since B_1 contains e'_3 and e'_2 , we have $v_3(B_1) \leq \max\{v_3(P_1), v_3(P_2)\} = 1$, which implies $v_3(B_2) = v_3(P_1 \cup P_2) - v_3(B_1) \geq 2 - 1 = 1$. $\square$

Then we allocate B_2 to agent 3 and do a re-partitioning of items in $\hat{M} = B_1 \cup P_3$ based on v_1 . Same as before, we have the guarantee that $v_1(\hat{M}) \geq 15/7$ and $v_2(\hat{M}) \geq 15/7$. However, a key difference is that now the most valuable item has been allocated to agent 3. As before, we run a *load-balancing* on items $\hat{M}$ and v_1 to get a partition $\{C_1, C_2\}$ with $v_1(C_1) \geq v_1(C_2)$.

CLAIM B.3. We have $v_1(C_2) \geq 6/7$.

PROOF. Suppose that $v_1(C_2) < 6/7$, we have $v_1(C_1) > 9/7$. Then by the same argument we have shown above, we have $|C_1| = 2$ and the most valuable item in $\hat{M}$ is allocated to C_2 . Since $e_1 \notin \hat{M}$, we have $v_1(C_1) \leq v_1(e_3 + e_4) \leq 1$, which is a contradiction. $\square$

Therefore, we have $v_1(C_1) \geq v_1(C_2) \geq 6/7$. Finally, we let agent 2 pick a bundle between $\{C_1, C_2\}$ first and agent 1 get the other bundle. The resulting allocation is $6/7$ -AMMS.